one was more realistic, with distances between points closer to the real distance in miles between cities, it could also generate some unnecessary confusion in the drawings; moreover, as we mentioned, the coordinate system is not really important for the algorithm, as long as it's consistent.

The first point we choose as pivot is "Opal City"[3]: since we first split using x coordinates, all cities on its left will go into one partition, and all cities on its right will go to the other partition. Then, as a second step, let's focus on the right partition: we choose, for instance, "Civic City" as a pivot, and this time we have to use y coordinates to split, so all points in the top-right region will go in a sub-partition of the right one, and all points in the bottom-right region will go into the other. We now have 3 partitions, and we can further split any of them.

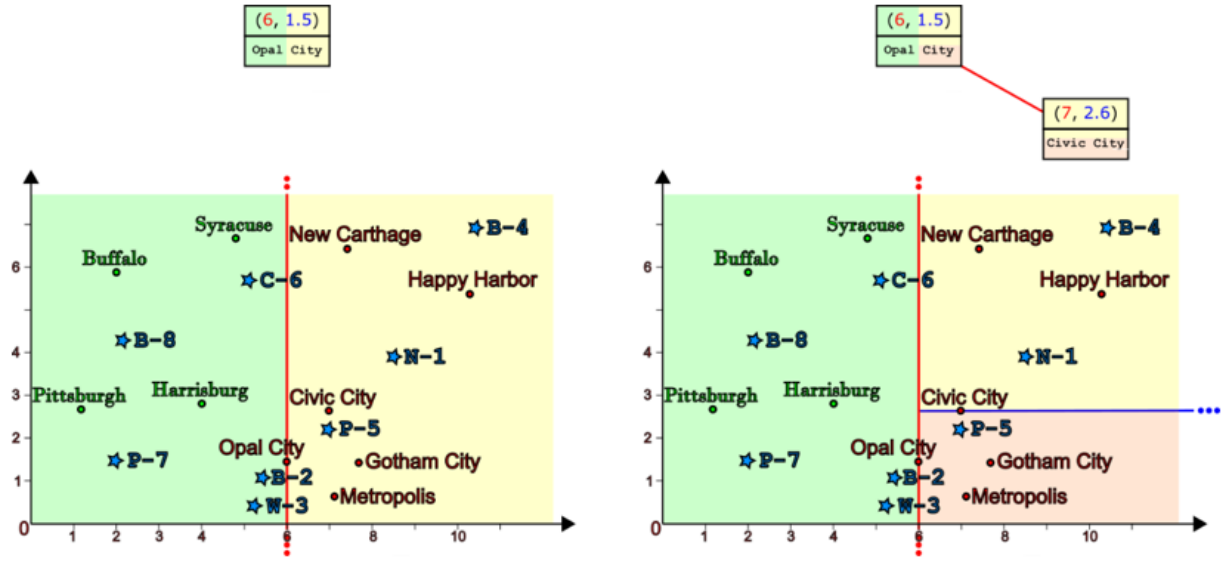


Figure 9.5 The first 2 steps of constructing a k-d tree for our map of DC cities.
(Left) First we split vertically with pivot "Opal City", which becomes the root of the k-d tree. Vertical splits